



Project Phase 2

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Introduction

In the rapidly evolving landscape of healthcare, integrating technology is crucial to meet the increasing demand for accessible, efficient, and personalized patient care. This project emerged from the need to adapt to changes brought about by the COVID-19 pandemic, heightened demand for healthcare services, and the swift advancement of technology. These factors highlighted the necessity of enhancing management systems, particularly customer relationship management (CRM) systems, to be more flexible, modern, efficient, and responsive.

This project aims to develop a comprehensive Hospital Customer Relationship Management (CRM) system tailored to the specific needs of a healthcare institution [1]. At this stage, our primary goal is to gather and understand the detailed requirements of the system's users, including patients, medical professionals, and hospital staff, and to seek additional information as needed to complete these requirements and identify key actors.

From customer insights and feedback, it is clear that there is a strong desire for an application that makes accessing healthcare services faster and more convenient, while also improving safety and the relationship between patients and the healthcare organization. To address these needs, we are developing a new, comprehensive application designed to enhance the efficiency of healthcare access and foster better patient engagement with the organization. Our approach begins with gathering and defining essential functional and non-functional requirements.

Functional requirements for the system include capabilities for remote consultations, automated reminders, secure telemedicine through video consultations, and medication schedule management. Additionally, the system will support secure payment processing for medical services, provide communication channels directly with healthcare providers, and facilitate medication delivery. The system will also include secure data storage, support multiple languages, and offer detailed reporting on user activity.

Non-functional requirements are also essential to ensure the system's overall effectiveness and user satisfaction. The CRM system must have a user-friendly interface to minimize learning curves and accommodate all users, including elderly or disabled patients. Accessibility features will include support for screen readers, adjustable text sizes, and voice

control. The system should be highly reliable, with all-day availability and minimal downtime, supported by a scalable cloud-based infrastructure. Cross-platform compatibility for both iOS and Android devices is necessary to provide a seamless user experience. Moreover, robust security measures are crucial for protecting sensitive patient data. The system must include multi-factor authentication, advanced encryption with AES-256, and regular security audits. To manage threats effectively, advanced security tools such as Endpoint Detection and Response (EDR) [2], Security Information and Event Management (SIEM) [3], and Security Orchestration, Automation, and Response (SOAR) will be implemented [4]. These tools will aid in detecting, analyzing, and responding to security incidents promptly. Additionally, the system must comply with key healthcare regulations, including the PDPA [5], Thai Medical Act [6], and HIPAA, to ensure legal and ethical handling of patient information.

In summary, the development of this Hospital CRM system is a strategic response to the evolving needs of the healthcare industry. By focusing on user-friendly design, robust security, and comprehensive functionality, this project aims to deliver a solution that enhances patient satisfaction, improves operational efficiency, and supports the digital transformation of healthcare services. All in all, these planning and defining the objectives and direction for this phase of the project will serve as the critical framework and structure for smooth transitioning into the next stages of the project.

Revision of the deliverables presented in the phase 1

Functional Requirements

FR1: The system shall allow patients to create a new account in the system to access services.

FR2: The system shall allow patients to update their account password for security purposes.

FR3: The system shall allow patients to create, reschedule, and cancel appointments based on recommendations or personal needs.

FR4: The system shall automatically send reminders to the patient about upcoming appointments.

FR5: The system shall allow patients to search for available doctors by specialty, schedule, or rating.

FR6: The system shall allow patients to view and manage their personal and medical data.

FR7: The system shall allow doctors to conduct a video consultation with the patient to provide advice and guidance

FR8: The system shall allow doctors to issue an electronic prescription for medication after consultation.

FR9: The system shall allow doctors to update the patient's medical records after consultations or treatments.

FR10: The system shall allow patients to view their medical history, including past consultations and treatments.

FR11: The system shall allow the patient to receive reminders about medication dosages according to the doctor's prescription.

FR12: The system shall allow the patient to input their symptoms into the system and receive preliminary advice based on symptom analysis.

FR13: The system shall allow the patients to receive tailored recommendations or advice based on their symptoms, after checking symptoms.

FR14: The system shall allow patients views, health tips, hospital updates, and other notifications sent by the system.

FR15: The system shall allow the patient to communicate with hospital support staff for assistance with issues or inquiries.

FR16: The system shall allow the patient to log into the system using secure authentication protocols.

FR17: The system shall facilitate payment processing for patients' medical services and medication purchases.

FR18: The system shall allow the patients to receive a detailed invoice after a service or medication purchase.

FR19: The system shall allow the system administrators to generate reports on system usage, appointments, and user activity.

FR20: The system should be able to create detailed invoices or bills for services rendered and medications dispensed.

FR21: The system shall allow the drug delivery service to deliver medications to patients after receiving a prescription from the doctor.

FR22: The system shall allow the patients to designate an emergency contact within their profile.

FR23: The system shall allow the patients to provide feedback on their experience with the system or services.

Non-functional Requirements

NFR1: The system must be available 24/7, utilizing a cloud-based architecture to ensure high availability and minimal downtime.

NFR2: The system should support up to 10,000 concurrent transactions. Implement load balancing techniques and closed-based architecture to manage the high user load. Consider using microservices for scalability.

NFR3: The system should be easy to use and not require extensive training. The user interface (UI) design must consider the needs of elderly patients, ensuring a seamless and intuitive user experience.

NFR4: The system should meet ISO 27001 standards for information security management.

NFR5: The system must protect patient data through multi-factor authentication (MFA) and encryption standards such as AES-256 for data at rest and in transit.

NFR6: The system must comply with relevant regulations, including the Personal Data Protection Act (PDPA) [5], the Thai Medical Act [6], and, if applicable, the Health Insurance Portability and Accountability Act (HIPAA) for international patients.

NFR7: Regular security audits and penetration testing should be conducted at least quarterly to identify and address vulnerabilities.

NFR8: The system should implement advanced security tools like Endpoint Detection and Response (EDR) [2], Security Information and Event Management (SIEM) [3], and Security Orchestration, Automation, and Response (SOAR) to manage threats effectively [4].

NFR9: The system should be designed with accessibility features, including support for color blindness.

NFR10: The system should include an ethical framework to ensure patient data is used responsibly and transparently.

NFR11: The system should be compatible with both iOS and Android operating systems, ensuring a consistent user experience across different devices.

Actors

Primary Actors

- **Patient/User:** The primary users of the system. They can create accounts, book appointments, manage their schedules, participate in telemedicine consultations, track medication schedules, and provide feedback on their experiences. This group includes patients with diverse needs, such as the elderly who require accessibility features (e.g., screen readers, voice commands, and color blindness support).
- **Doctor:** Healthcare providers who use the system to conduct video consultations, issue prescriptions, update patient medical records, and manage patient care remotely. They are responsible for delivering treatment and advice, accessing patient data securely, and communicating with patients through the system.

Supporting Actors

- **System Administrator:** Responsible for managing the technical infrastructure of the system, ensuring it is available 24/7. They oversee security protocols, manage system performance (including load balancing and scalability), and conduct regular security audits and penetration testing. They ensure compliance with data protection regulations and standards such as the PDPA (Personal Data Protection Act) [5] and ISO 27001.
- **Hospital/Clinic Staff:** Support personnel who assist patients with inquiries, manage appointment bookings, and facilitate communication between patients and doctors. They help ensure a seamless user experience and handle issues related to patient data or system functionality, ensuring smooth system operations.
- **Security Manager:** A specialized role responsible for ensuring the system's security measures are properly implemented. They manage security tools such as Endpoint Detection and Response (EDR) [2], Security Information and Event Management (SIEM) [3], and Security Orchestration, Automation, and Response (SOAR) [4]. They are responsible for encryption standards, multi-factor authentication (MFA), and compliance with regulations such as HIP thAA,e PDPA [5], and the Thai Medical Act [6].
- **Payment Gateway:** Facilitates secure financial transactions for medical services and medication purchases. It ensures that payments are processed securely using encryption

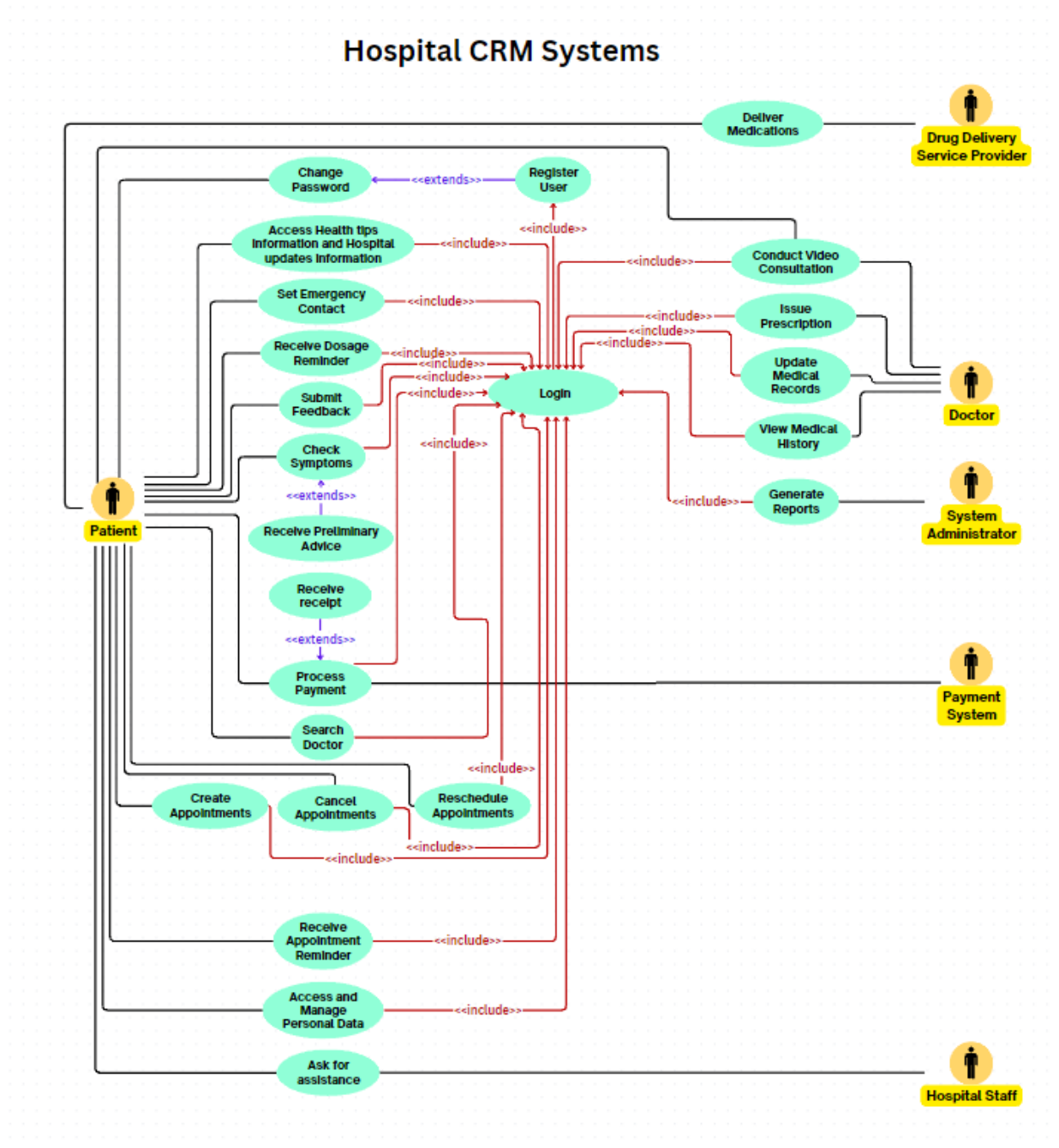
and compliance with legal and financial standards, such as PCI-DSS, to protect patient financial data.

- **Credit Card Processors:** These systems handle the secure processing of credit card payments for medical services and medications. They work alongside payment gateways to ensure that credit card transactions are completed securely and comply with financial standards, including encryption and fraud prevention protocols.
- **Mobile Banking Systems:** These systems integrate with mobile banking applications to allow patients to pay for services and medications securely through their mobile devices. They ensure compliance with security measures, including encryption and multi-factor authentication (MFA), to protect patient financial information during transactions.
- **Regulatory Authority:** External entities ensuring that the system complies with local and international regulations, such as the Personal Data Protection Act (PDPA), the Thai Medical Act [6], and HIPAA (Health Insurance Portability and Accountability Act). They may require periodic audits, certifications (e.g., ISO 27001), and compliance reporting.
- **Cloud Service Provider:** Provides the cloud-based infrastructure that supports the system's high availability, scalability, and performance. They manage concurrent transactions, ensuring the system remains operational 24/7 with minimal downtime. Cloud service providers are critical for maintaining system uptime and reliability.

Updated Actors

- **Patient:** Interacts with the system to manage appointments, consult doctors, receive reminders, etc.
- **Doctor:** Provides consultations, issues prescriptions, and interacts with patient records.
- **Hospital Staff:** Supports patient inquiries, manages promotions, and maintains operations.
- **Pharmacy/Delivery Service:** Delivers prescribed medications to patients (external service).
- **Payment System (External):** Processes secure payments, including credit card and mobile banking transactions.
- **System Administrator:** The system administrator generates reports on system usage, appointments, and user activity.

Use Case Diagram



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Description for the diagram

1. Register User

- **Actors:** Patient, Doctor, Drug Delivery Service Provider, Hospital Staff
- **Description:** Allows new users, including patients, doctors, delivery personnel, and hospital staff, to create an account in the system, granting them access to services and functionalities based on their roles.

2. Change Password

- **Actors:** Patient
- **Description:** Enables patients to securely update their account password to enhance security.

3. Create Appointments

- **Actors:** Patient
- **Description:** Allows patients to schedule appointments with doctors based on availability and personal needs.

4. Cancel Appointments

- **Actors:** Patient
- **Description:** Provides patients the ability to cancel scheduled appointments if no longer needed or as per medical advice.

5. Reschedule Appointments

- **Actors:** Patient
- **Description:** Enables patients to modify the date and time of an existing appointment based on availability or personal preference.

6. Receive Appointment Reminder

- **Actors:** Patient
- **Description:** Sends automated reminders to patients about upcoming appointments to ensure attendance.

7. Search Doctor

- **Actors:** Patient
- **Description:** Allows patients to search for doctors based on specialty, availability, or ratings to find the most suitable match.

8. Access and Manage Personal Data

- **Actors:** Patient
- **Description:** Provides patients with the ability to view and update their personal and medical information in the system.

9. Conduct Video Consultation

- **Actors:** Doctor, Patient
- **Description:** Facilitates a virtual consultation between doctors and patients for medical advice and diagnosis.

10. Issue Prescription

- **Actors:** Doctor
- **Description:** Allows doctors to create and issue electronic prescriptions for patients after a consultation.

11. Update Medical Records

- **Actors:** Doctor
- **Description:** Enables doctors to add or update information in a patient's medical record after consultations or treatments.

12. View Medical History

- **Actors:** Patient
- **Description:** Allows patients to view their complete medical history, including previous consultations, treatments, and prescriptions.

13. Receive Dosage Reminder

- **Actors:** Patient
- **Description:** Provides automated reminders to patients for medication dosages based on the doctor's prescription.

14. Check Symptoms

- **Actors:** Patient
- **Description:** Allows patients to input symptoms for analysis and receive preliminary advice on potential conditions or next steps.

15. Receive Preliminary Advice

- **Actors:** Patient

- **Description:** Offers patients initial guidance based on symptom input, with suggestions for possible next steps, like seeing a doctor.

16. Access Health Information and Notifications

- **Actors:** Patient
- **Description:** Enables patients to view general health tips, hospital updates, and notifications provided by the system.

17. Communicate with Support Staff

- **Actors:** Patient, Hospital Staff
- **Description:** Allows patients to contact hospital support staff for assistance with issues or inquiries related to the system.

18. Login

- **Actors:** Patient, Doctor, Drug Delivery Service Provider, Hospital Staff
- **Description:** Allows all registered users, including patients, doctors, delivery providers, and hospital staff, to access the system using their account credentials.

19. Process Payment

- **Actors:** Patient, Payment System
- **Description:** Processes payments for medical consultations and medication purchases, including billing and transaction confirmations.

20. Receive receipt

- **Actors:** Patient
- **Description:** Provides a detailed receipt to patients for medical services or medication purchases.

21. Generate Reports

- **Actors:** System Administrator
- **Description:** Allows the system administrator to generate analytical reports on system usage, appointment statistics, and user activities.

22. Deliver Medications

- **Actors:** Drug Delivery Service Provider, Patient
- **Description:** Coordinates with the delivery service to send prescribed medications to patients' designated addresses after receiving an electronic prescription.

23. Set Emergency Contact

- **Actors:** Patient
- **Description:** Allows patients to designate and update an emergency contact within their profile for medical emergencies.

24. Submit Feedback

- **Actors:** Patient
- **Description:** Enables patients to submit feedback on their experiences with the system and medical services received, helping improve service quality.

Use Case Narrative

Use Case Name	Make Remote Appointments
Goal in Context	Enable patients to schedule, reschedule, or cancel medical appointments remotely through the Hospital CRM system.
Primary Actor	Patient
Secondary Actor	• Doctor • Hospital Staff
Precondition	• The patient must be registered and logged into the system. • Available appointment slots must exist for the selected doctor or clinic.
Trigger	The patient initiates the appointment booking process by selecting the "Make Appointment" option on the CRM system.
Scenario	1. The patient logs into the Hospital CRM system. 2. The patient navigates to the "Make an Appointment" section. 3. The system displays a list of available doctors and appointment slots. 4. The patient selects a doctor based on specialization and availability. 5. The system prompts the patient to choose a date and time for the appointment. 6. The patient confirms the chosen date and time. 7. The system verifies the availability of the selected slot. 8. The system sends an appointment confirmation notification to the patient via email/SMS. 9. The doctor and relevant hospital staff are notified of the new appointment. 10. The system sets reminders for the appointment (e.g., 24 hours and 1 hour prior).
Exceptions	1. No available time slots – The system prompts the patient to select a different date or doctor. 2. Incomplete patient information – The system requires the patient to update their profile or medical history before confirming the appointment. 3. System error during appointment booking – The system displays an error message and prompts the user to retry.
Post-condition	• The appointment is successfully booked and confirmed. • Notifications and reminders are sent to both the patient and the doctor. • The appointment is added to the patient's and doctor's schedules in the system.

Use Case Name	Receive Appointment Reminders
Goal in Context	Ensure patients receive timely reminders for their scheduled appointments to reduce missed consultations.
Primary Actor	Patient
Secondary Actor	• Doctor • Hospital Staff
Precondition	• The patient has a confirmed appointment (remote or in-person) scheduled in the system. • The patient has provided contact information (email, phone number, or preferred communication method).
Trigger	The scheduled appointment time is approaching (e.g., 24 hours before the appointment).
Scenario	1. The system checks for upcoming appointments within the predefined reminder period (e.g., 24 hours and 1 hour before). 2. The system sends an automated reminder to the patient via the patient's preferred communication channel (email, SMS, or in-app notification). 3. The system provides details such as the appointment date, time, doctor's name, and any instructions (e.g., preparing for a remote consultation). 4. The system also sends a reminder to the doctor about the scheduled appointment. 5. If the patient confirms or cancels via the reminder message, the system updates the appointment status accordingly. 6. The system logs the reminder notification in the patient's record.
Exceptions	1. Invalid contact details – If the patient's contact information is outdated or incorrect, the system flags it and prompts the patient to update their contact information. 2. Patient does not receive the reminder (e.g., due to email or SMS server issues) – The system attempts to resend the reminder or notify hospital staff if delivery repeatedly fails.
Post-condition	• The patient is reminded of their upcoming appointment, reducing the likelihood of missed consultations. • The system logs the reminder event, and any confirmations or changes are updated in the patient's record.

Use Case Name	Manage Appointment Schedules
Goal in Context	Allow patients to manage (view, reschedule, or cancel) their appointments efficiently through the hospital CRM system.
Primary Actor	Patient
Secondary Actor	• Doctor • Hospital Staff
Precondition	• The patient must have an active account in the system. • The patient has scheduled at least one upcoming appointment.
Trigger	The patient logs into the system and accesses their appointment schedule.
Scenario	1. The patient logs into the hospital CRM system. 2. The patient navigates to the "Appointments" section, where they can view a list of scheduled appointments. 3. The patient selects an appointment they wish to manage. 4. The system provides options to: ○ Reschedule: The patient selects a new date and time from the available options, and the system checks for the doctor's availability. ○ Cancel: The patient confirms the cancellation, and the system updates the doctor's schedule. 5. The system updates the appointment status (rescheduled or canceled) and notifies the doctor of the change. 6. If the appointment is rescheduled, the system sends a confirmation message with the new details to the patient. 7. The system logs the action in the patient's record.
Exceptions	1. No available slots – If the patient attempts to reschedule but no slots are available, the system provides alternative options or prompts the patient to contact hospital staff. 2. System outage – If the system is unavailable, the patient is advised to contact hospital staff for manual appointment management. 3. Deadline exceeded – If the patient tries to cancel an appointment after the allowed time window (e.g., within 24 hours of the appointment), the system blocks the action and prompts the patient to contact hospital staff.
Post-condition	• The patient's appointment schedule is updated according to their changes (rescheduled or canceled).

	• The system notifies the doctor of any changes to their schedule.
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Use Case Name	Process Secure Payments
Goal in Context	Allow patients to securely pay for medical services and medications via the CRM system, supporting both credit card and mobile banking payment options.
Primary Actor	Patient
Secondary Actor	• Payment Gateway • Hospital Staff • System Administrator
Precondition	• The patient has received medical services or purchased medication, and an invoice has been generated. • The patient has an active account in the CRM system.
Trigger	The patient receives an invoice and initiates the payment process.
Scenario	1. The patient logs into the CRM system. 2. The system displays the patient's outstanding invoice(s). 3. The patient selects "Make Payment" and chooses a payment method (credit card or mobile banking). 4. The system connects to the secure payment gateway. 5. The patient enters payment details (credit card or mobile banking credentials). 6. The payment gateway processes the transaction. 7. The system verifies the payment and updates the hospital's billing records. 8. The system sends a payment confirmation to the patient via email or notification. 9. The hospital staff is notified of the successful payment.
Exceptions	• If the payment is declined or fails, the system notifies the patient and prompts them to try another payment method. • If the payment gateway is down, the system notifies the patient of a delay and attempts to retry the transaction later. • If there is insufficient balance in the patient's account, the system alerts the patient to retry when funds are available.

Post-condition	• The payment is processed securely, and the patient receives a receipt • The hospital's billing system is updated with the completed transaction.
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Use Case Name	Update Patient Records Post-Consultation
Goal in Context	Allow doctors to update and manage patient records after consultations, ensuring accurate and up-to-date medical information.
Primary Actor	Doctor
Secondary Actor	• Patient • System Administrator • Hospital Staff
Precondition	• The doctor has completed a consultation with the patient. • The patient's medical records exist in the system.
Trigger	The doctor needs to update the patient's medical records after a consultation.
Scenario	1. The doctor logs into the CRM system. 2. The doctor selects the patient's medical record. 3. The doctor updates the medical notes, diagnosis, or treatment plan. 4. The doctor saves the updated information, and the system securely stores the new data. 5. The system notifies the patient that their medical records have been updated. 6. The patient can access their updated records through the CRM system.
Exceptions	1. If the doctor's session times out due to inactivity, the system prompts the doctor to log in again and resume the update. 2. If the system detects a data conflict (e.g., simultaneous updates by hospital staff), it prompts the doctor to review the changes before saving. 3. If the system encounters a failure in saving the updates, it retries automatically and logs an alert for the system administrator.
Post-condition	• The patient's medical records are updated and securely stored in the system. • The patient is notified of the updates and can access the new

	information.
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Update Use Case Narrative

Since there are several new case diagrams being revised, I would like to update the descriptions of the new use cases.

Use Case Name	Create Appointments
Goal in Context	Enable patients to schedule medical appointments with doctors through the hospital CRM system.
Primary Actor	Patient
Secondary Actor	
Precondition	• The patient must be registered, logged into the system, and have an active account.
Trigger	The patient initiates the appointment booking process by selecting the "Create Appointment" option in the CRM system.
Scenario	1. The patient logs into the CRM system and navigates to the "Patient Dashboard" section. 2. The patient navigates to the "Create Appointment" section. 3. The system displays available doctors, dates, and time slots. 4. The patient selects a doctor based on specialization and an available time slot. 5. The system verifies the availability of the chosen slot and confirms the appointment. 6. The system sends an appointment confirmation notification to the patient and the doctor. 7. The system schedules a reminder for the appointment closer to the scheduled time.
Exceptions	• No available slots: If no slots are available, the system prompts the patient to select a different time or doctor. • Incomplete patient information: The system requires the patient to complete any missing profile or medical details before booking. • System error: The system displays an error if any issues occur during booking and asks the patient to retry.
Post-condition	• The appointment is successfully booked, confirmed, and added to both the patient's and doctor's schedules in the system.

Use Case Name	Ask for Assistance
Goal in Context	Allow patients and hospital staff to request assistance from the support team or other hospital staff.
Primary Actor	Patient, Hospital Staff
Secondary Actor	
Precondition	The user must be logged into the system.
Trigger	The user selects the "Ask for Assistance" option within the CRM system.
Scenario	- The user navigates to the "Ask for Assistance" section. - The system displays a form where the user can describe their issue or request. - The user fills out the form with the necessary details and submits the request. - The system logs the request and assigns it to the appropriate support team or hospital staff. - The support team receives a notification about the new assistance request. - The support team contacts the user to provide assistance.
Exceptions	Incomplete request details: The system prompts the user if essential details are missing. Support team unavailable: The system notifies the user if support is delayed due to unavailability.
Post-condition	The assistance request is logged, and the support team follows up to resolve the issue.

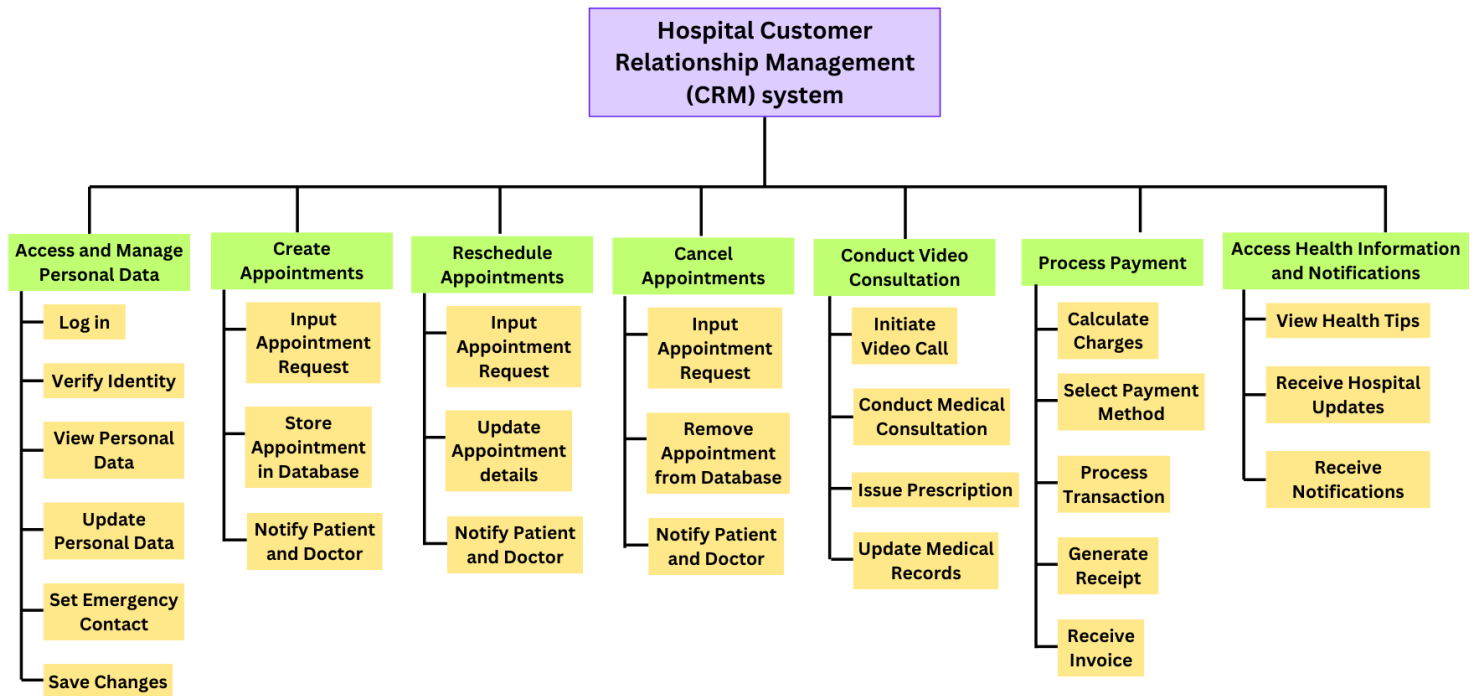
Use Case Name	Deliver Medications
Goal in Context	Enable the Drug Delivery Service Provider to deliver prescribed medications to patients.
Primary Actor	Drug Delivery Service Provider
Secondary Actor	
Precondition	The patient has received a prescription that is approved for delivery.
Trigger	A prescription order is created and flagged for delivery in the CRM system.
Scenario	1. The Drug Delivery Service Provider receives a notification for a new delivery request. 2. The system provides the delivery provider with the necessary prescription details and the patient's contact and delivery information. 3. The delivery provider prepares the medication for dispatch. 4. The medication is dispatched to the patient's address. 5. The system updates the delivery status to "In Transit." 6. Upon successful delivery, the provider marks the order as "Delivered" in the system.
Exceptions	Invalid delivery address: The system notifies the provider if the address is incorrect, and the provider contacts the patient for verification. Medication out of stock: If the medication is unavailable, the system notifies the doctor and patient to consider alternatives.
Post-condition	The medication is successfully delivered to the patient, and the delivery status is updated in the system.

Use Case Name	Update Medical Records
Goal in Context	Allow doctors to update patient medical records after consultations, ensuring accurate and up-to-date information.
Primary Actor	Doctor
Secondary Actor	
Precondition	The doctor has completed a consultation with the patient, and the patient's records are accessible.
Trigger	The doctor initiates a medical record update for the patient after a consultation.
Scenario	- The doctor logs into the CRM system and accesses the patient's medical records. - The doctor updates relevant details, such as diagnosis, treatment plans, or medication prescriptions. - The doctor saves the updates, and the system securely stores the new information. - The system notifies the patient that their medical records have been updated. - The patient can view the updated records through their account.
Exceptions	Session timeout: If the doctor's session times out, the system prompts them to log in again. Data conflict: If another user is updating the records simultaneously, the system alerts the doctor to review before saving. Save error: If there is a save error, the system retries and alerts the System Administrator if necessary.
Post-condition	The patient's medical records are updated and securely stored, and the patient is notified of the changes.

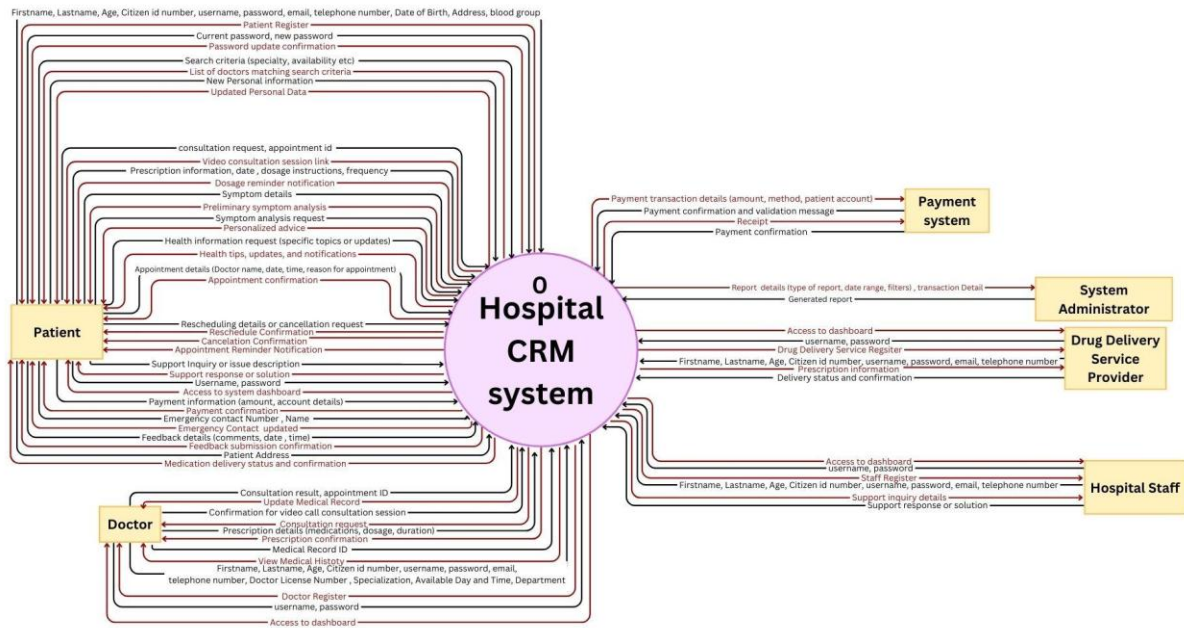
Use Case Name	Generate Reports
Goal in Context	Allow System Administrators or relevant hospital staff to generate various reports, such as appointment statistics, financial summaries, or patient records.
Primary Actor	System Administrator
Secondary Actor	
Precondition	The user must have the necessary permissions to access and generate reports.
Trigger	The user initiates the report generation process within the CRM system.
Scenario	- The user logs into the CRM system and navigates to the "Generate Reports" section. - The system displays available report options (e.g., appointments, billing, medical records). - The user selects a report type and applies relevant filters (e.g., date range, department). - The system gathers the required data based on the selected parameters. - The system compiles the data into the selected format (e.g., PDF, Excel) and generates the report. - The report is made available for download or viewing on the system.
Exceptions	Insufficient permissions: If the user lacks necessary permissions, the system restricts access and notifies them. Data retrieval error: If there is an error retrieving data, the system displays an error message and suggests retrying later.
Post-condition	The report is successfully generated and available for download or viewing.

Functional decomposition diagram

(Structure chart)



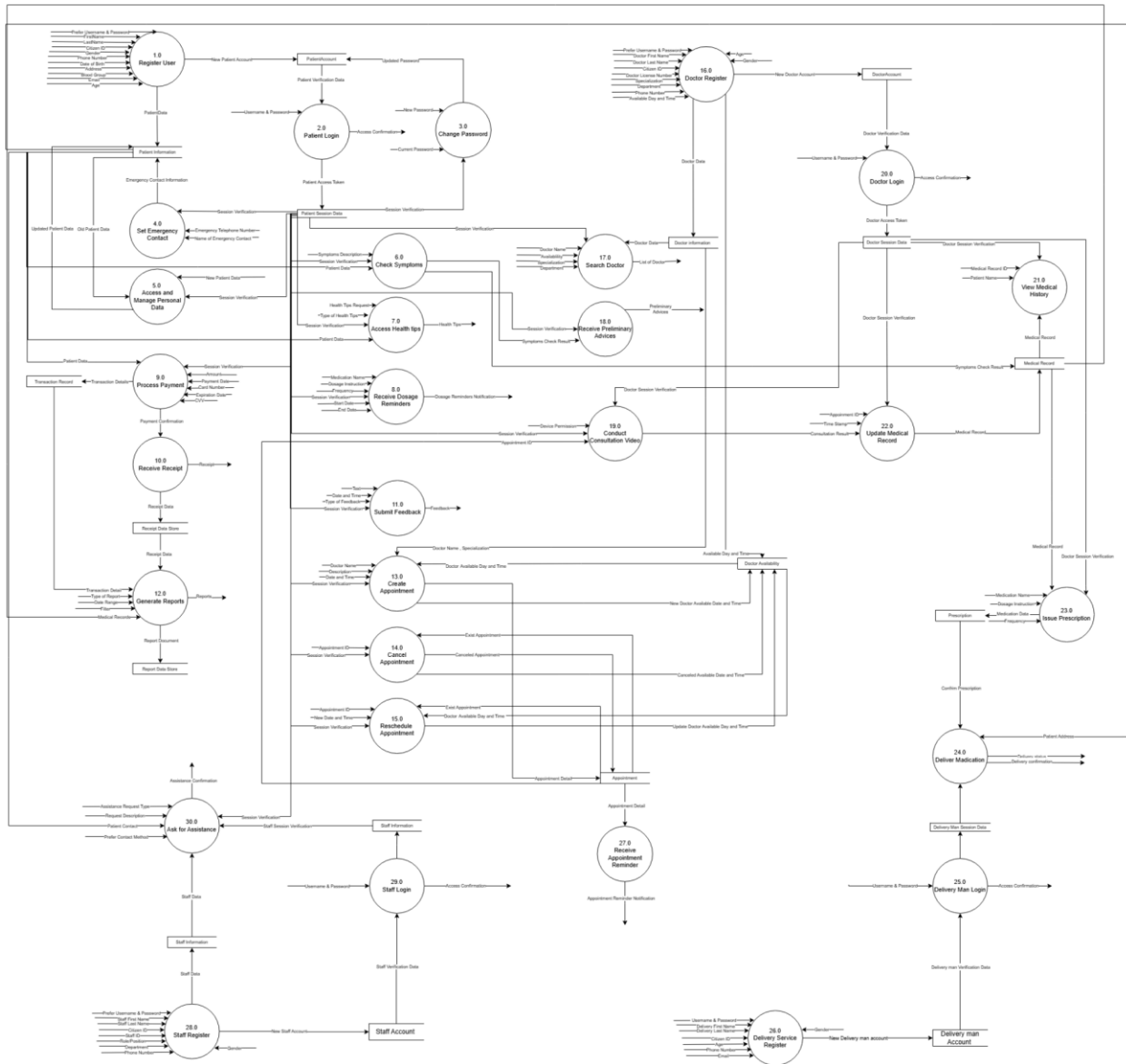
Context Diagram (DFD Level 0)



Link:

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DFD Level 1

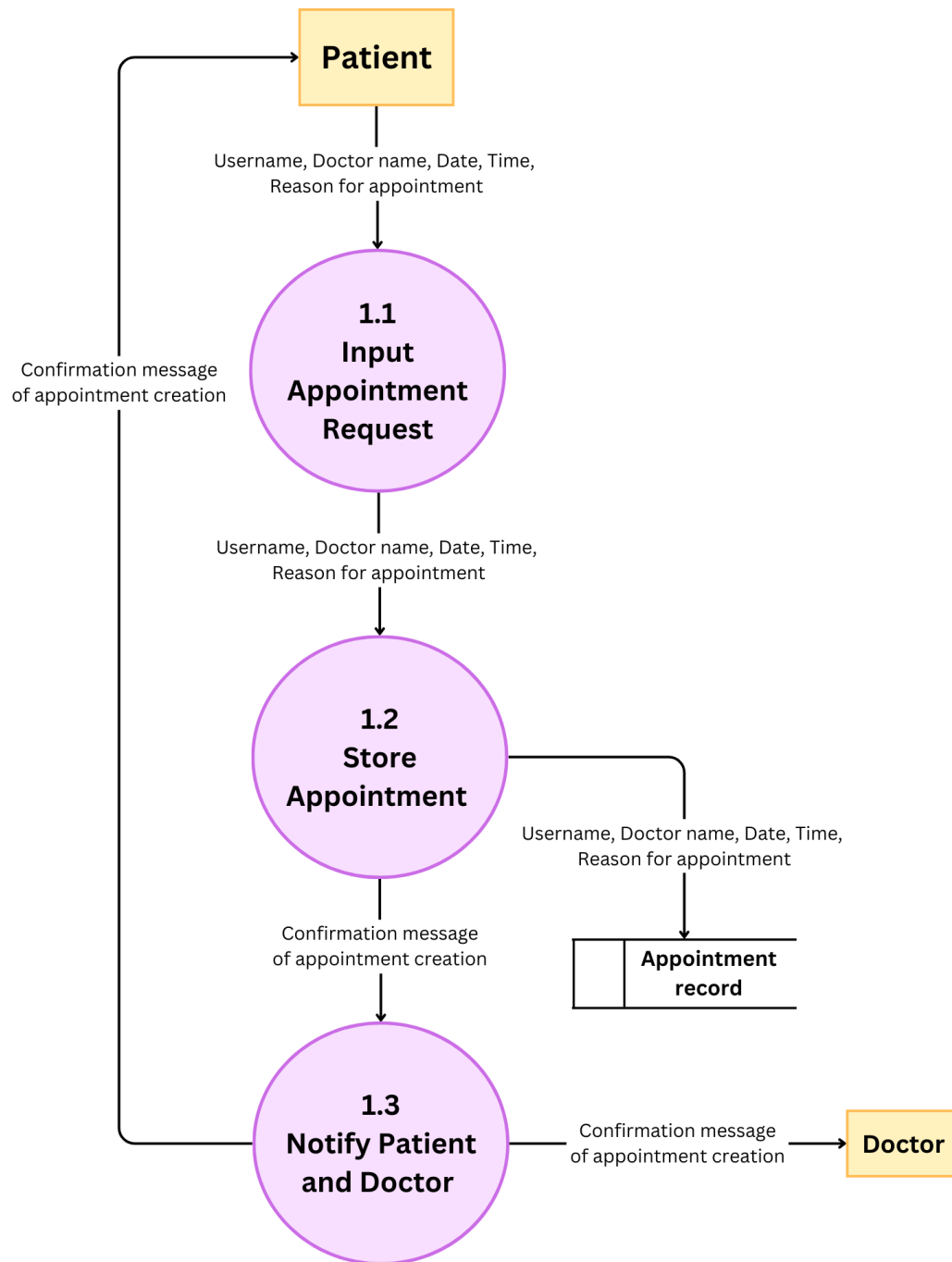


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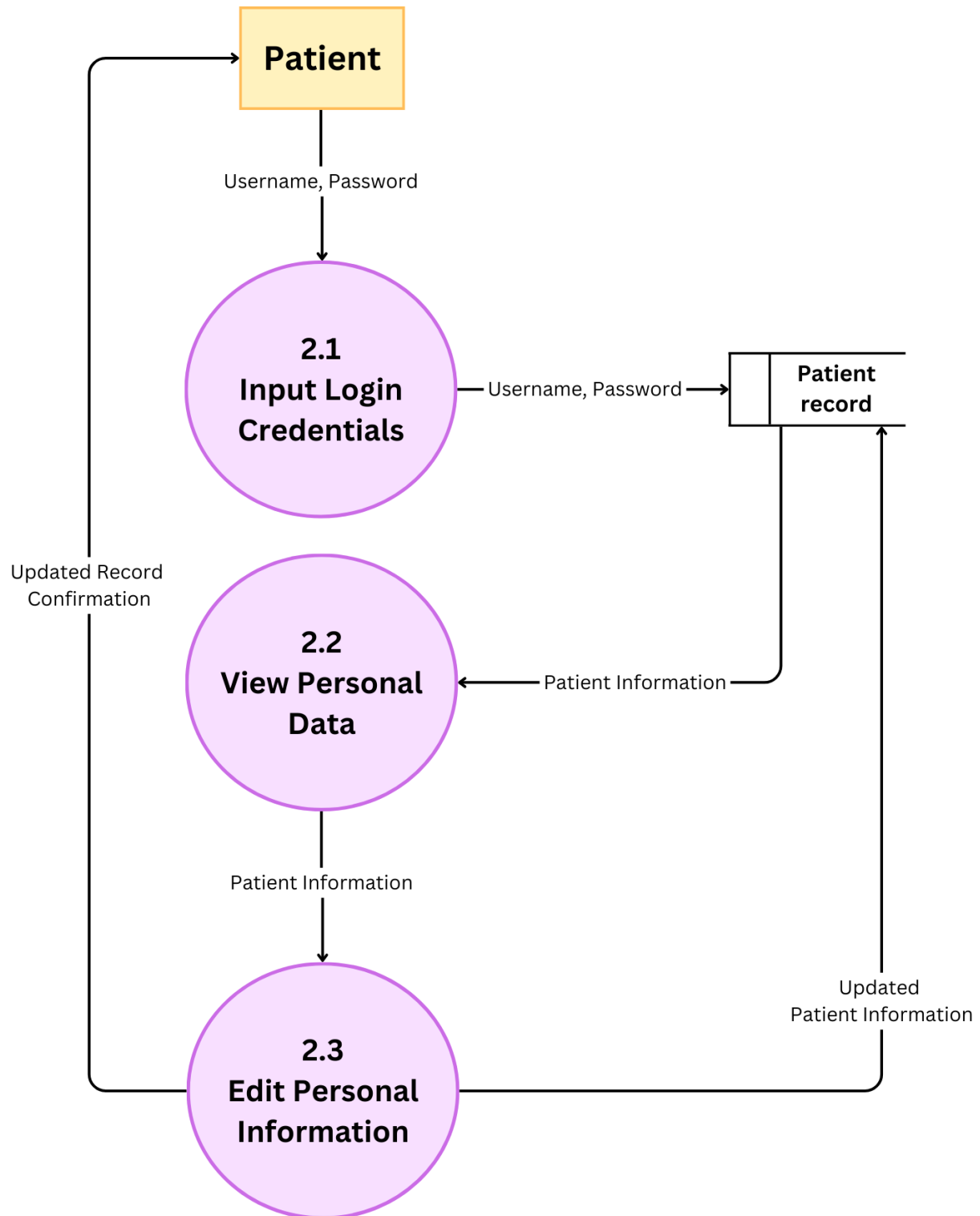
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Data Flow Diagram Level 2

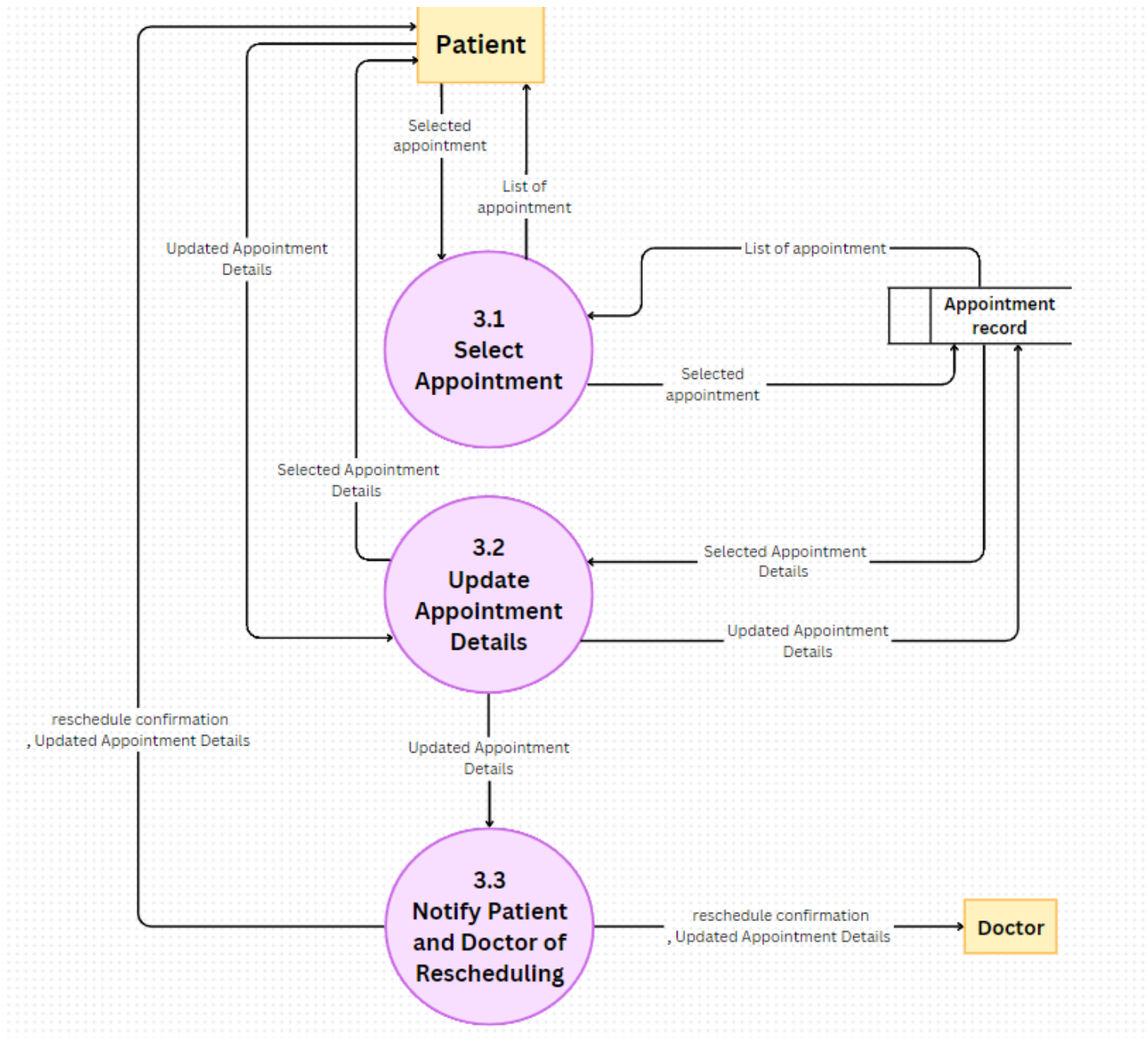
Create Appointment



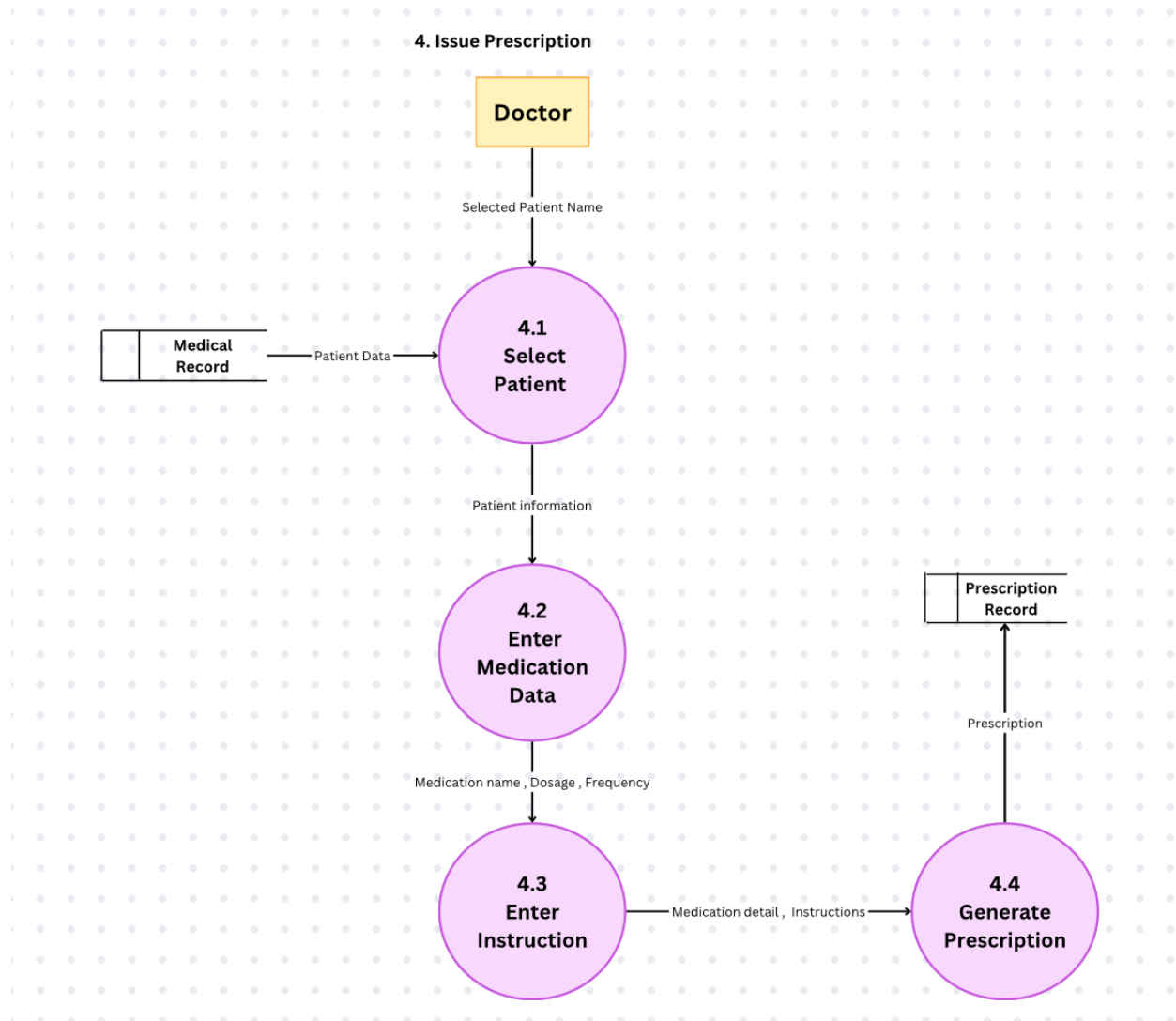
Access and Manage Personal Data



Reschedule Appointment



Issue Prescription



Working prototype

HOSPITAL CRM SYSTEM

A comprehensive system for hospital management

[Home](#) [Our Team](#)



Patient Mode

Book appointments, view medical records, and communicate with doctors.



Doctor Mode

Access patient records, manage schedules, and update treatment plans.



Admin Mode

Manage users, oversee system settings, and generate reports.

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HOSPITAL CRM SYSTEM

Please login to continue

Login



notesuppakorn

☐ Patient ☐ Doctor ☐ Admin

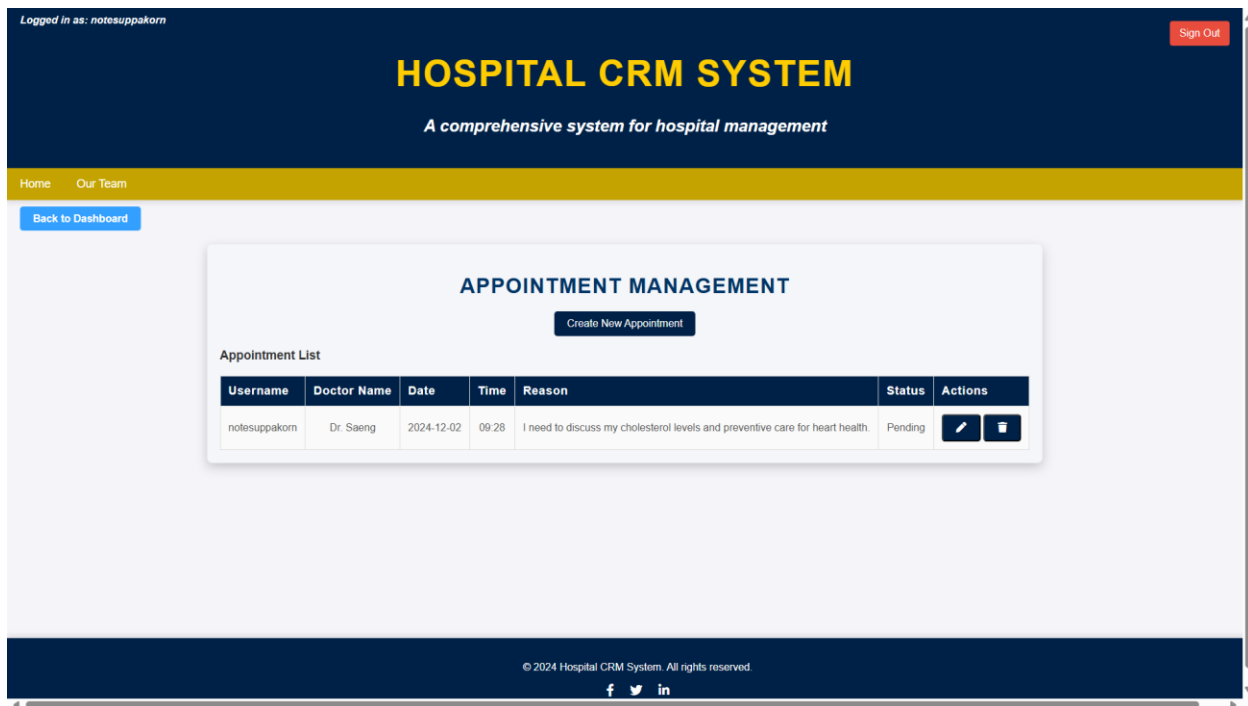
Login

Back to Home

Register

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HOSPITAL CRM SYSTEM

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Book Appointment

Logged in as: notesuppakorn

Select Doctor

Dr. Saeng (Cardiology)

Appointment Time

mm/dd/yyyy --:--

Reason

Book Appointment

Cancel

Logged in as: notesuppakorn

Sign Out

HOSPITAL CRM SYSTEM

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Home Our Team

Back to Dashboard

APPOINTMENT MANAGEMENT

Create New Appointment

Appointment List

Username	Doctor Name	Date	Time	Reason	Status	Actions
notesuppakorn	Dr. Saeng	2024-12-02	09:28	I need to discuss my cholesterol levels and preventive care for heart health.	Pending	 

Edit Appointment

Doctor Name Dr. Saeng (Cardiology) Date 12/02/2024 Time 09:28 AM Reason I need to discuss my cholesterol Save Changes Cancel

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ACCOUNT INFORMATION

 Username: notesuppakorn
 Password: *****
 First Name: Note
 Last Name: Suppakorn
 Citizen ID: 123456789
 Gender: Male
 Phone Number: +1234567890
 Date of Birth: 1990-01-01
 Address: 123 Main St, City
 Blood Group: O+
 Email: notesup@example.com
 Age: 34

[Back to Patient Dashboard](#)[Edit Information](#)

Edit Account Information

First Name

Last Name

Citizen ID

Gender

Phone Number

Date of Birth

Address

Blood Group

Email

Age

[Save Changes](#)[Cancel](#)

Logged in as: docsaeng

Sign Out

HOSPITAL CRM SYSTEM

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[Home](#) [Our Team](#)

Doctor Dashboard

Manage Appointments

Check and manage your patient's appointments.

View Patient History

View your patient's medical records and history.

Update Medical Record

Edit or update patient medical records.

Issue Prescription

Issue new prescriptions for your patients.

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Logged in as: docsaeng

Sign Out

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[Back to Dashboard](#)

Issue Prescription

Select Patient

Note Suppakorn

Medication Name

Enter medication name

Dosage

Enter dosage details (e.g., 500mg)

Frequency of Medication

Select a frequency

Special Instructions

Enter any special instructions here

Issue Prescription

[Back to Dashboard](#)

Issue Prescription

Select Patient

Note Suppakorn

Medication Name

Enter medication name

Dosage

Enter dosage details (e.g. , 500mg)

Frequency of Medication

Select a frequency

Special Instructions

Enter any special instructions here

Issue Prescription

Patient	Medication	Dosage	Frequency	Instructions	Action
Note Suppakorn	paracetamon	1000MG	threeTimesADay	Don't take it with alcohol.	 

Test cases

1. Test Case for Create Appointment:

Test Case ID: Create_Appointment_01

Test Priority: High

Module Name: Appointment Management

Test Title: Verify creation of appointment process

Description: Test that the create appointment page can process the appointment creation, stores the appointment data, and sends notifications to both the patient and doctor.

Test Case Designed by: Paranee Wannasorn

Test Designed Date: 10 November 2024

Test Executed by: Suppakorn Pojsomphong

Test Execution Date: 15 November 2024

Pre-conditions:

- 1) The patient is logged into the website with a valid account.
- 2) The doctor is available and listed on the website.

Dependencies:

- 1) The doctor's availability schedule is accurate and up-to-date.

Step No.	Test Step	Test Data	Expected Result	Actual Result	Status (Pass/Fail)
1.	Click on "Create New Appointment" button	N/A	Appointment creation form opens successfully	Appointment creation form loaded successfully.	Pass
2.	Select a doctor from the dropdown	Doctor: "Dr. Saeng (Cardiology)"	Doctor "Dr. Saeng (Cardiology)" is displayed in the selection	Doctor "Dr. Saeng (Cardiology)" displayed correctly.	Pass
3.	Choose an appointment date and time	Date: "14/11/2024" Time: "14:30"	Selected date and time are shown in the appointment form	Date and time displayed correctly in the form.	Pass
4.	Click "Book Appointment"	N/A	Appointment is confirmed, and confirmation message "Appointment with Dr. Saeng confirmed for 2024-11-14T14:30" is displayed	Confirmation message: "Appointment with Dr. Saeng confirmed for 2024-11-14T14:30"	Pass
5.	Check if appointment appears in the list	Username: "notesuppakorn" Doctor: "Dr. Saeng" Date: "2024-11-14" Time: "14:30"	Appointment details are displayed in the appointment list with "Edit" and "Delete" options	Appointment details listed successfully with "Edit" and "Delete" options.	Pass

2. Test Case for Access and Manage Personal Data:

Test Case ID: Access_PersonalData_02

Test Priority: High

Module Name: Account Management

Test Title: Verify Access and Update of Personal Data

Description: Test that the user can access and update personal data in the account management module.

Test Case Designed by: Ratchabhol Songserm

Test Designed Date: 10 November 2024

Test Executed by: Yotsawan Sangkavaree

Test Execution Date: 15 November 2024

Pre-conditions:

- 1.) The user is logged into the system with a valid account.
- 2.) The user has navigated to the Account Management section from the dashboard.

Dependencies:

- 1) The user's personal data is available and stored in the system.

Step No.	Test Step	Test Data	Expected Result	Actual Result	Status (Pass/Fail)
1.	Click on "Account Management" button	N/A	Account Management page is displayed successfully	Account Management page displayed successfully.	Pass
2.	Access and view personal data	Username: "notesupakorn"	User's personal details (name, contact, etc.) are displayed correctly	Personal details displayed correctly.	Pass
3.	Edit personal data fields (e.g., phone, address)	Phone: "0810184523" Address: "Bangkok"	Updated details are accepted and shown correctly in the form	Updated details displayed correctly.	Pass
4.	Click "Save" to update data	N/A	System saves the updated data	Personal information saved successfully.	Pass
5.	Verify the updated personal data is saved	N/A	The system shows the confirming message that the data updated correctly.	Confirmation message displayed: "Details updated successfully."	Pass

3. Test Case for Reschedule Appointment

Test Case ID: Reschedule_03

Test Priority: High

Module Name: Appointment Management

Test Title: Verify Rescheduling of Appointment

Description: Test that the appointment management page allows the patient to reschedule an existing appointment, updates the appointment details, and notifies both the patient and doctor correctly.

Test Case Designed by: Paranee Wannasorn

Test Designed Date: 10 November 2024

Test Executed by: Yotsawan Sangkavaree

Test Execution Date: 15 November 2024

Pre-conditions:

- 1) The patient is logged in and has at least one existing appointment.
- 2) The user has navigated to the Appointment Management section from the dashboard.

Dependencies:

- 1) The appointment records in the system are stored correctly.

Step No.	Test Step	Test Data	Expected Result	Actual Result	Status (Pass/Fail)
1.	Click on “Appointment Management” button	N/A	Appointment list opens successfully	Appointment list opened successfully.	Pass
2.	Select an existing appointment to reschedule	Doctor: "Dr. Saeng" Date: "2024-11-14" Time: "14:30"	Selected appointment is displayed with all current details (Doctor name, Date, Time).	Selected appointment displayed correctly.	Pass
3.	Click on "Edit"	N/A	Edit appointment form opens successfully	Edit form opened successfully.	Pass
4.	Update the appointment details (Date, Time, Reason)	New Doctor Name: "Dr. Nai" New Date: "2024-11-25" New Time: "11:30" Reason: "Follow-up Consultation"	The system displays the updated appointment details (Doctor name, Date, Time, Reason).	Updated details displayed correctly in the form.	Pass
5.	Click on "Save Changes"	Updated Appointment Details Doctor Name: "Dr. Nai" Date: "2024-11-25" Time: "11:30"	Appointment is updated in the system and appointment details are stored.	Updated details saved successfully.	Pass

6.	Verify the system updates appointment record	Updated Details Updated Details: Dr. Nai, 2024-11-25, 11:30	Appointment record is updated with new details.	System record updated correctly with new details.	Pass
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4. Test Case for Issue Prescription

Test Case ID: Issue_Prescription_04

Test Priority: High

Module Name: Issue Prescription

Test Title: Verify Issue Prescription

Description: Verify the functionality to issue, edit, and delete prescriptions

Test Case Designed by: Paranee Wannasorn

Test Designed Date: 10 November 2024

Test Executed by: Yotsawan Sangkavaree

Test Execution Date: 15 November 2024

Pre-conditions:

- 3) The doctor is logged into the system using a valid account.
- 4) The user has navigated to the “Doctor Mode” section from the dashboard.

Dependencies:

- 2) The prescription records in the system are stored correctly.

Step No.	Test Step	Test Data	Expected Result	Actual Result	Status (Pass/Fail)
1.	Click on “Doctor Mode”	N/A	Doctor Mode section opens successfully	Doctor Mode section opens successfully	Pass
2.	Login as a doctor	Username: "docsaeng" Password: "inwza123"	Doctor is authenticated and redirected to the Doctor Dashboard	Doctor is authenticated and redirected to the Doctor Dashboard	Pass
3.	Click on "Issue Prescription"	N/A	Issue Prescription page is displayed	Issue Prescription page is displayed	Pass
4.	Fill in prescription details	Patient Name: "Note Suppakorn" Medication Name: "Amoxicillin" Dosage: "500 mg" Frequency: "Once a day" Special Instructions: "Take with food to avoid stomach upset"	Prescription details are entered successfully	Prescription details are entered successfully	Pass
5.	Click "Issue Prescription"	N/A	Prescription is saved to the table below to show the entered data and the doctor can also edit or delete the issue prescription.	Prescription is saved to the table below to show the entered data and the doctor can also edit or delete the issue prescription.	Pass

References

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